

DATA STRUCTURE

LAB PROGRAM

Write a C program to arrange a series of numbers using Quick Sort

```
#include <stdio.h>
```

```
void quicksort(int a[], int low, int high)
```

```
{
```

```
    int i, j, pivot, temp;
```

```
    if(low < high)
```

```
    {
```

```
        pivot = a[low];
```

```
        i = low + 1;
```

```
        j = high;
```

```
        while(i <= j)
```

```
        {
```

```
            while(a[i] < pivot && i <= high)
```

```
                i++;
```

```
            while(a[j] > pivot)
```

j--;

if(i < j)

{

temp = a[i];

a[i] = a[j];

a[j] = temp;

}

}

temp = a[low];

a[low] = a[j];

a[j] = temp;

quicksort(a, low, j - 1);

quicksort(a, j + 1, high);

}

}

int main()

{

int a[100], n, i;

```
printf("Enter number of elements: ");
scanf("%d", &n);

printf("Enter elements:\n");
for(i = 0; i < n; i++)
    scanf("%d", &a[i]);

quicksort(a, 0, n - 1);

printf("Sorted array:\n");
for(i = 0; i < n; i++)
    printf("%d ", a[i]);

return 0;
}
```

Output

```
Enter number of elements: 4
```

```
Enter elements:
```

```
46 64 99 2
```

```
Sorted array:
```

```
2 46 64 99
```

```
=== Code Execution Successful ===
```